

atomOS v3.6

Kernel validation & texture residency

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Native CUDA 12.8 / sm_120. Local synthetic K1 inputs. This report replaces the earlier partial-residency report and distinguishes the final binary from retained exploratory builds.

RESULT	OBSERVED EVIDENCE
Full 4 MiB dictionary	18/18 maximum-atlas cold profiles reached exactly 131,072 compulsory misses. The dictionary is distributed across 46 SM caches.
After-warm retention	24/24 final cold profiles reached their dictionary miss floor across default and additional shape/seed cases.
Full computation retained	212 bulk-path conformance runs; 26 complete exported traces independently checked by Python.
Memory and synchronization	Final bulk binary: six memchecks, four syncchecks and four racechecks. The rebuilt production binary passed six additional memchecks.
Claim boundary	Measured retention through the verified cooperative warm/work/reread epoch. No guaranteed cache pinning across launches or arbitrary competing workloads.

What changed

The optional bulk-copy kernel stages live inputs and results through native asynchronous shared/global transfers while masks stay on the texture path. Correctly rounded constants replace repeated sine/cosine calls for the fixed .4 rotation. The final kernel uses 84 registers per thread and has zero stack/local allocation.

Residency has an execution cost. The report includes all warming, probing, barriers and diagnostic work in ordinary kernel timing. A complete-residency observation is not a claim of the fastest possible K1 implementation.

What the cache result means

The measurement starts cold, reads the entire physical dictionary, runs the real K1 epoch, and rereads the entire dictionary. Grid-wide barriers separate the phases. Inputs are immutable throughout. The host independently checks every candidate before committing word and JK state.

PHASE	COVERAGE / CHECK
Warm	Every packed uint4 texel, including physical padding. Each aligned 32-byte sector is required at least once.
Real work	All logical lanes execute the original producer, word disposition, JK, blend, OTAN2 and six diagnostic checks. Padding executes no logical work.
Post-work reread	Every physical texel again, on its owning cooperative block. Independent expected per-thread XOR digests cover all texels, but can collide.
Ownership	Exactly one active 512-thread block per SM is required. All 46 observed SM identifiers must be unique and unchanged at both endpoints of every launch.

At 4 MiB, $4,194,304 / 32 = 131,072$ unique texture sectors. A cold launch with exactly that many misses leaves no observed extra dictionary misses for work or the final reread. This inference depends on the recorded cold-cache and coverage conditions; the counters are aggregate, not separate measurements of a 100% hit rate in each phase. [1]

SIZE / LAYOUT	MISS FLOOR	MEASURED MISS RANGE	REPEATS AT FLOOR
default / linear	2,048	2,048 - 2,048	3/3
default / morton8	2,048	2,048 - 2,048	3/3
max / linear	131,072	131,072 - 131,072	3/3
max / morton8	131,072	131,072 - 131,072	3/3

Nsight Compute: kernel replay, cache-control all, clock-control none, setup launch skipped, one measured launch per sample. Requested sectors, hits, misses, realized shared configuration, global-load traffic and profiler duration are retained with raw locale-aware CSV values. Missing metrics never become zero.

Cache operators and carveout preferences are hints, not a supported texture-line pinning interface. Passing this experiment does not establish retention across a new launch or guarantee behavior on another GPU. [2]

Native transfer schedule and repairs

The full dictionary is divided into aligned physical intervals. A 64-texel tile stages 64 Lane records plus 64 State records in shared memory. Once all readers have private values, the same shared union holds 64 complete Result records. Native bulk-copy completion and proxy fences protect every reuse. [3, 4]

CHANGE	WHY IT MATTERS
Native bulk I/O	cp.async.bulk moves Lane/State into shared memory and Results back to global memory. Mask reads remain tex1Dfetch<uint4>. Diagnostic exports also use bulk copies.
Fixed rotation folded exactly	cos(binary64 .4) = 0x1.d7954e7dba2f8p-1; sin(binary64 .4) = 0x1.8ec3ae92b676bp-2. The original CPU/GPU literal and runtime calls matched both constants bit for bit.
Exact rounding evidence	40-term rational alternating Taylor sums and next-term bounds lie strictly inside the nearest-double rounding bins. The high-precision calculation and GPU value probe are separate evidence categories.
Padded backing and canonical export	Padded row-major Lane/State/Result buffers keep bulk ranges aligned. Morton physical tiles map explicitly to these records. Zero output padding is verified before canonical logical rows are gathered.
Verify before commit	Downloaded candidates, checksums, SM records and padding must all pass. Exported traces contain actual GPU results and pre-commit states; incomplete output remains in an uncommitted staging directory.

The controlled padded before/after comparison changed only the fixed rotation constants in the computation: maximum-atlas excess misses fell from 508 to zero in the first sample of each layout. Registers also fell from 107 to 84. This supports the compiled change; it does not isolate coefficient traffic as the sole microarchitectural cause.

Literal atan(delta_phase / delta_rho), separately selected atan2 completion, whole-word absorption, explicit J/K, separate shift-XOR/shift-OR, unavailable statuses, Morton bit order and original comparison thresholds remain intact. The nearby rotated-singularity regression preserves the difference between literal and directed profiles.

Machine code and memory footprint

FINAL BULK KERNEL	MEASURED / INSPECTED VALUE
Device / architecture	NVIDIA GeForce RTX 5070 Ti Laptop GPU / compute capability 12.0
Toolchain	CUDA 12.8.61; MSVC 19.44.35221 (toolset 14.44.35207); CMake 4.3.2; native sm_120 plus compute_120 PTX
Registers / stack / local	84 per thread / 0 bytes / 0 bytes
Shared allocation	10,752-byte shared union + 8-byte barrier = 10,760 user bytes. cuobjdump reports 11,784 bytes including the 1,024-byte difference.
Realized shared configuration	16,384 bytes in every final counter sample; maximum-L1 preference is recorded separately.
Native texture / bulk instructions	15 static TLD sites, 18 UBLKCP.S.G sites and 13 UBLKCP.G.S sites. Loop/unrolled sites are not dynamic traffic counts.
Remaining global bookkeeping	Zero ordinary LDG/STG in the inspected kernel body. Two generic load sites and two atomic sites support cooperative barriers; profiler global-load traffic is still nonzero.

At the atlas cap, 262,144 uint4 texels occupy 4 MiB. On this 46-SM GPU each block owns at most 5,760 texels, or 90 KiB. The 4 MiB is distributed across the SMs. It is not 4 MiB resident in each SM, and maximum-L1 is not evidence of an exclusively available texture-cache capacity. [2]

ALLOCATION	ACTUAL FORMULA
Device payload	272 x stored_texels + 16 x SMs x (2 x 512 + 2) bytes
Diagnostics on this GPU	755,136 bytes for pre/post thread digests and two uint4 SM records per block
Admission	Payload + 64 MiB margin; 512 MiB ceiling; preserve 1,536 MiB of actual free VRAM; retain the 70%-of-free limit

The final exporter binary and the pre-export folded binary have identical kernel SASS and encoded instruction words. Native code inspection supports instruction-path claims; it does not constitute a proof of the compiled binary. Source FMA contraction is disabled; explicit fused instructions used inside device math libraries may still occur.

Correctness and execution coverage

EVIDENCE CATEGORY	EXECUTED RESULT
C++ contract regression checks	32 groups / 134,521 assertions; includes fixed-rotation singularity and nearby available cases
Independent Python regressions	37 tests; includes rejected fabricated bulk profiles, incorrect memory accounting and incomplete receipts
CPU execution	7 CTest cases; 48 independently verified exported configurations
Symbolic specification	15 Z3 4.16.0 obligations: every negated property UNSAT. These are mathematical specifications, not a C++/CUDA binary proof.
Rebuilt production CUDA	144 normal configurations; 612,000 lane-epochs; six additional memchecks
Final bulk conformance	192 main + 20 additional aspect/seed cases; every candidate checked before commit
Independent bulk trace exports	26 runs / 626,288 lane-epochs recomputed by Python; run seals checked
Bulk Compute Sanitizer	6 memcheck + 4 synccheck + 4 racecheck runs; zero reported errors or race hazards/warnings

The 192-case main bulk matrix covers 1x1, 17x257, 128x1024 and 512x16384 shapes; both layouts; all four producers; fringe on/off; source, directed and mixed diagnostic profiles; three committed epochs. Selected exports cover every producer and profile, with both maximum-atlas layouts exported for an additional one-epoch check.

The additional 20 cases cover 1x65536, 32768x1, 128x65536, 511x16353 and 9x33, both layouts, and seeds 0 / 4,294,967,295 with different producer/profile/fringe choices. Memory, synchronization and race checks include both padded-tail and full maximum fixtures.

Largest independent floating discrepancy: production 8.88e-16 rad; bulk exports 8.88e-16 rad. Acceptance stays 2e-11 rad; the invariant threshold stays 1e-10. Integer fields, statuses, states and reasons compare exactly. CPU address/undefined-behavior sanitizers were not rerun in this Windows review.

Retention, traffic and execution cost

SIZE / LAYOUT	MEDIAN	TRIAL RANGE	SCOPE
default / linear	76.16 us	71.10 - 81.34 us	warm + work + probe
default / morton8	83.20 us	76.58 - 84.54 us	warm + work + probe
max / linear	2470.40 us	2470.37 - 2472.38 us	warm + work + probe
max / morton8	2482.82 us	2480.67 - 2769.34 us	warm + work + probe

Each timing row contains five separate, shuffled processes, each with three measured epochs after a verified, uncommitted setup launch. The statistic is the median of each process's epoch median. Upload, download, host verification and export are outside the CUDA-event interval. Nsight and sanitizer timings are excluded from this table.

FINAL COLD COUNTERS	SECTORS (MEDIAN)	MISSES (MEDIAN)	HITS (MEDIAN)
default / linear	6,154	2,048	4,106
default / morton8	6,156	2,048	4,108
max / linear	393,750	131,072	262,678
max / morton8	393,750	131,072	262,678

The minimum nominal traffic for a full unpadded dictionary is three reads: warm, productive work and final probe. Aggregate sectors can exceed that nominal count; all excess requests are retained in the study. A higher hit percentage alone would not demonstrate an improvement, because redundant reads enlarge the denominator.

Earlier controls explain the search, not the final performance claim

Global warming alone did not remove productive TEX misses. With split I/O, no-allocation stores preserved the large dictionary in the store-only control, but input loads did not. Native bulk I/O reduced the full 4 MiB case to 508 excess misses; folding the fixed rotation removed those remaining observed misses in the controlled comparison.

The earlier production sweep found a 33.1% lower best median after launch tuning (81.25 to 54.34 us at the default shape). That measurement belongs to the archived pre-fold binary and excludes warm/probe work. It is retained as historical evidence, not relabelled as a benchmark of the final bulk binary. Complete retention and minimum runtime are separate objectives.

Reproduce and use the delivered kernel

The delivered `atomos_cache_bulk.exe` implements the full measured schedule and supports `--out` for independent verification. Keep its profile explicit: `bulk-warm-work-probe-v1` with `bulk-padded-row-major-v1` allocation. The standard `atomos_cuda.exe` retains its texture, packed-texture and global choices.

```
cmake -S . -B C:/Users/Tom/.cache/ak1/bulk_export_sm120
-T cuda=12.8 -DATOMOS_ENABLE_CUDA=ON
-DATOMOS_CACHE_BULK_EXPERIMENT=ON
cmake --build C:/Users/Tom/.cache/ak1/bulk_export_sm120
--config Release --target atomos_cache_bulk
```

Enter each wrapped command above as one command. The short build directory avoids the reproduced Windows MSBuild path-length issue; the explicit installed toolset avoids stale CUDA 12.9 integration.

```
output/bin/atomos_cache_bulk.exe --rows 512 --angles 16384
--layout morton8 --mode recurrent --epochs 1 --out my_bulk_run
python tools/verify_run.py my_bulk_run
```

The output directory must be new. Canonical run files contain actual downloaded GPU candidates, with no logical padding rows. The Python verifier checks the complete trace, final state, exact execution profile and actual allocation formula. Seals record local file integrity; they are not author authentication.

```
python tools/validate.py --proofs
python tools/validate.py --gpu --sanitizer --proofs
--build C:/Users/Tom/.cache/ak1/gpu_fold_final
--cmake-arg=-Tcuda=12.8
```

Use `tools/study_cache_bulk.py` for the 226-command main study and `tools/check_cache_bulk_edges.py` for the 46-command supplementary study, passing the executable, a new evidence directory, and the native Nsight/Compute Sanitizer paths recorded in their JSON. Both retain raw logs, before/after source and binary hashes, and an independent receipt audit.

Existing repairs remain: allocation admission preserves the reserve in currently free VRAM; inconsistent execution metadata is rejected; Windows sanitizer discovery resolves the native executable. No drivers, watchdog settings, privileges or service connections were changed.

Evidence, identity and limits

All paths below are local. K = C:/Users/Tom/.cache/ak1. R is the workspace root printed below. Each study records exact commands, device metadata, raw logs and executable/source hashes. The final source manifest and output delivery manifest are unsigned integrity records.

C:\OMT AC031 Womack remarkable carmack oculus master quest half-life 3 gabecubelatomOS_v3_6_RTX5070Ti_Laptop_Kernel_Tom_KlootwijklatomOS_v3_6_CUDA_Verification

EVIDENCE	LOCATION
Final review / current status	R/results/residency_followup/summary.json; R/results/validation_status.json
CPU validation	K/cpu_fold_final/evidence_1789299038566882500/validation.json
Production GPU validation	K/gpu_fold_final/evidence_1789299027401576500/validation.json
Main bulk study	K/bulk_final_study_v2/study.json; raw .stdout.log/.stderr.log/.csv.log files and sealed *.run directories alongside it
Additional shapes / sanitizers	K/bulk_final_edges/study.json; raw per-command logs alongside it
Rounding / native code	R/results/residency_followup/rotation_rounding.json; rotation_fold_native.json; final_native.json; bulk_final_sass.txt
Initial controlled cache comparison	R/results/residency_followup/padded_before_fold_profiles.json; fold_initial_profiles.json
Retained rejected attempt	K/bulk_final_study/study.json: validator-source change during execution; excluded from final counts

Final bulk executable SHA-256

fc25b2f435bc3c1ccbcddc2ed8f494ef3247285b357541b72f2855e8b83a175e4

Final production executable SHA-256

cdfe79ac3e48e35f7837954b680ad9bbef0a2a4ba353422e54724ff6bed75d95

Limits of the conclusion

The final profile set records 24 miss-floor successes and 0 excess-miss outcomes. These are finite, workload-specific hardware observations. They do not establish cache pinning, arbitrary concurrency, all possible inputs, every future compiler, or a formal CUDA-binary proof. K1 remains the synthetic integrated word-epoch profile; unimplemented engine producers remain listed in docs/coverage.csv.

Primary references

- [1] Nsight Compute: cache control and 32-byte L1/TEX sector metrics. Full URLs and claim mapping: docs/cache_sources.json.
- [2] PTX ISA 8.7: cache hints, memory model and texture coherence. Full URLs and claim mapping: docs/cache_sources.json.
- [3] CUDA 12.8: bulk asynchronous transfers and completion ordering. Full URLs and claim mapping: docs/cache_sources.json.
- [4] PTX ISA 8.7: cp.async.bulk and async-proxy fences. Full URLs and claim mapping: docs/cache_sources.json.